

A LiDAR-Based Recommendation System to Improve Sitting Posture Respecting Privacy

Francesco Pistolesi^{ID}, *Member, IEEE*, Michele Baldassini^{ID}, and Beatrice Lazzerini^{ID}, *Member, IEEE*

Abstract—Poor posture is becoming more widespread due to the rising number of jobs that require workers to sit for extended hours. Maintaining proper leg positioning is essential for good overall posture and long-term health. However, current monitoring methods involve multiple sensors and cameras, leading to discomfort and privacy concerns. This paper presents a privacy-preserving recommendation system using Light Detection and Ranging (LiDAR) to monitor leg positions. The system captures the horizontal leg outline at knee height and extracts domain-specific features, while a stacked autoencoder reproduces the latent representation of the leg outline. The system comprises three modules, including one-class and multi-class support vector machines, to identify 15 leg positions and reject unrecognized ones. Data from 30 healthy volunteers during work activities trained the system. When tested on new participants, the system achieved an accuracy of over 98%. In addition to monitoring leg positions while respecting privacy and ergonomics, our system alerts workers about poor leg positions and generates intuitive dashboards with posture statistics that help safety engineers identify at-risk workers and body parts. This offers a promising solution for improving desk workers' posture and reducing long-term health issues, respecting privacy.

Index Terms—Artificial intelligence, autoencoder, ergonomics, industry 5.0, LiDAR, occupational health, pattern recognition, posture, privacy, support vector machine.

I. INTRODUCTION

POSTURE is key to preventing many health problems. There has been a high increase in people sitting at computers for long periods in recent years [1]. Poor posture has become common, causing health issues as it stresses joints and muscles, leading to fatigue and overwork [2]. This may evolve into chronic pain and musculoskeletal disorders, potentially

Manuscript received 11 April 2023; revised 22 December 2023 and 27 March 2024; accepted 8 May 2024. Date of publication 22 May 2024; date of current version 6 September 2024. This work was supported in part by the Italian Ministry of University and Research (MUR) in the frameworks of the *CrossLab* and *FoReLab* projects (Departments of Excellence), and in part by the PNRR-M4C2-Investimento 1.3, Partenariato Esteso PE00000013-FAIR—Future Artificial Intelligence Research-Spoke 1 Human-Centered AI." (Corresponding author: Francesco Pistolesi.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by University of Pisa Bioethical Committee, Application No. 23/2022.

The authors are with the Department of Information Engineering, University of Pisa, 56122 Pisa, Italy (e-mail: francesco.pistolesi@unipi.it; michele.baldassini@ing.unipi.it; beatrice.lazzerini@unipi.it).

Digital Object Identifier 10.1109/JBHI.2024.3404233

damaging muscles, tendons, and ligaments [3]. After COVID-19 restrictions have been eased, people have started mixing working at home and in the office, increasing productivity by up to 13% [4]. But home working may lead to incorrect positions.

Ergonomics aims to develop environments that minimize the risk of injury, harm, or long-term negative consequences to health. Various solutions were recently proposed that promote ergonomics, particularly for desk workers. For example, a system was developed to encourage regular posture changes, also suggesting neck exercises [5]. Passive flexible sensors were integrated into seat pads to recognize posture using artificial neural networks implemented on field-programmable gate arrays [6]. Innovative chairs equipped with pressure sensors were developed to recognize posture based on processing rules [7], fuzzy rule-based algorithms [8], k-nearest neighbors classifiers [9], decision trees [10], and spiking neural networks [11]. Advanced chairs were also proposed that combined pressure and distance sensors [12], also to monitor the posture of wheelchair users [13], [14]. The Microsoft KinectTM was used for risk evaluation of posture [15] and to classify the types of 3D posture [16]. Other solutions used computer vision, with smart cameras and dynamic hierarchical hidden semi-Markov models [17]. On-body real-time location system tags are recent solutions to locate workers and detect behaviors [18].

Despite their good performance, sensor-based solutions typically use various devices on the chair and/or body, which may be annoying. Instead, computer vision solutions require video recording or capturing pictures, causing potential psychological discomfort to workers and putting privacy at high risk.

Posture monitoring techniques should be unobtrusive and respect privacy. The light detection and ranging (LiDAR) is one of those techniques. It generates a point cloud that accurately outlines the scanned environment, respecting privacy. We recently presented an embedded system that could distinguish six leg positions at the desk using 2D LiDAR [19]. That system used statistical features to represent a 2D LiDAR scan of the legs, associating it with one of six leg positions using a multi-class machine learning classifier. The technology preserves privacy as each scan is just a horizontal contour of the legs that cannot reveal sensitive information. Despite remarkable accuracy (99.2%), the system in [19] had limitations. For example, rotating the chair might reduce accuracy. Also, it could only recognize six leg positions, returning a random position for workers seated with a leg position outside those six.

This paper presents an advanced version that uses 2D LiDAR and artificial intelligence to recognize 15 sitting leg positions.

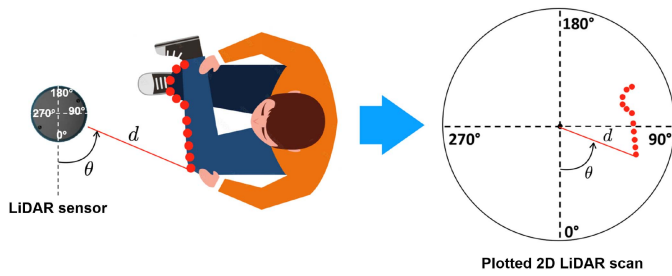


Fig. 1. 2D LiDAR sensor (left) and plot of a LiDAR scan (right).

A LiDAR sensor in front of the knees emits laser pulses and measures the distance they travel before hitting the legs: recorded distance values form a *LiDAR scan*. The system creates a sample combining statistical features extracted from the scan and part of the latent space of a stacked autoencoder trained to reproduce the scan. Two one-class classifiers (*O-C Good* and *O-C Bad*) and a multi-class classifier based on support vector machines cooperate to determine the leg position. *O-C Good* detects if the scan corresponds to a *correct leg position*: if it is, the classification task ends. If not, *O-C Bad* detects if the scan is an incorrect position that the multi-class classifier can recognize. If it is, the multi-class classifier detects the *incorrect leg position*; otherwise, the scan is an *anomaly* (unrecognized position), and the classification task is invalidated. The novelties compared to [19] are that the system:

- 1) recognizes triple the number of leg positions, thus enabling more accurate and specific risk analyses as each leg position stresses different body parts;
- 2) includes anomaly detection for unrecognized positions;
- 3) generates more advanced posture reports highlighting the workers at risk and the most stressed body parts;
- 4) sends real-time on-screen recommendations to workers to help them correct posture.

A total of 30 healthy people participated in an experiment during their work activities. Results showed over 98% accuracy in detecting leg positions.

The paper is structured as follows: Section II contains a background; Section III describes the data collection; Section IV details each module of the systems and how they were implemented; Section V compares the system to other classifiers; Section VI provides a demonstration of how the system works; Section VII draws the conclusions.

II. BACKGROUND

A. Light Detection and Ranging (LiDAR)

A 2D LiDAR sensor [20] (see Fig. 1) sends laser pulses as it rotates and measures the distance traveled by those returning after hitting something in the sensor range. It saves an *angle-distance pair* (θ, d) for each pulse, where θ is the angle between the pulse and a zero-degree reference point, whereas d is the distance of the hit object from the sensor. Pulses that do not return are assigned $d = 0$. Angle-distance pairs recorded during a 360-degree rotation form a *LiDAR scan*.

B. Autoencoders

An autoencoder (AE) is a neural network that represents an input $\mathbf{x} \in \mathbb{R}^n$ as a vector $\mathbf{c} \in \mathbb{R}^m$ in a lower-dimensional space (*latent-space representation*), where $m \ll n$, and generates an output $\mathbf{y} \in \mathbb{R}^n$ that is a reconstruction of $\mathbf{x}$ [21].

An AE has an *encoder* that creates the latent-space representation, and a *decoder* that uses this representation to get the output. The mean squared error $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2$ generally compares the output $\mathbf{y}$ to the input $\mathbf{x}$: the lower the MSE, the better. Autoencoders can be layered one after the other, obtaining a *stacked autoencoder*.

C. Support Vector Machine (SVM)

Classical SVMs split data into two classes by maximizing the margin between a hyperplane and the closest data points of each class in the feature space.

1) *One-Class SVM*: One-class SVMs are used when normal data are well-defined and anomalous data are not. This type of SVM does not require a training set of anomalous data, which can sometimes be difficult to obtain. There are two algorithms to implement this classifier aimed at finding a boundary separating the normal data from anomalous data.

a) *One-class SVM (O-C SVM)*: The O-C SVM [22] uses a kernel function to map the data into a high-dimensional space. It then searches for the hyperplane that maximizes the margin around the normal data points while minimizing the number of data points outside the margin.

b) *Support vector data description (SVDD)*: SVDD [23] uses a kernel function to find the smallest hypersphere containing normal data points. SVDD helps handle non-linear data, but may be more computationally intensive.

2) *Multi-Class SVM*: Multi-class SVMs [24] divide data into n classes, where $n > 2$. They consist of two types.

a) *One-vs-one*: The *one-vs-one* type uses a set of hyperplanes ($n/2$), each separating a pair of classes. The SVM then assigns a new data point to one class based on which side of each hyperplane the point falls on.

b) *One-vs-all*: The *one-vs-all* type trains n binary classifiers, each separating one class from the rest. To classify a new data point, the SVM selects the class whose hyperplane has the farthest distance from the point.

III. DATA COLLECTION

A. Participants

Data were collected from 30 healthy desk workers whose legs were filmed and LiDAR scanned after signing an informed consent form. This form outlined the experiment's purpose, the anonymity of data collection, and the deletion of videos after labeling. Participants were 15 men and 15 women aged 25 to 50, with 10 wearing skirts and the rest wearing trousers. Their weight ranged from 51.2 to 99.7 kg, and their height from 1.51 to 1.94 m.

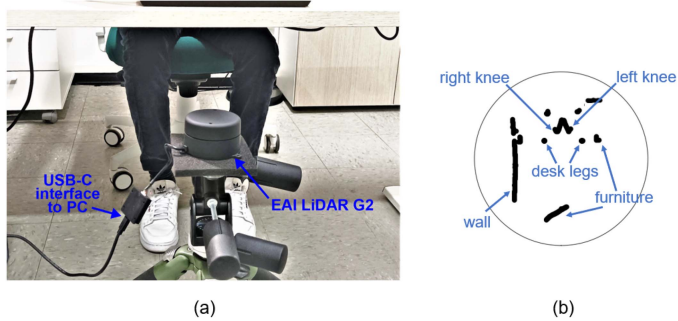


Fig. 2. Front view of a worker sitting with correct leg position (a) and plot of the corresponding LiDAR scan (b).

B. Sensor

An EAI YDLiDAR G2 collected 2D LiDAR scans of the legs that were sent to a PC via its serial port for storage. This sensor has a diameter of 7.37 cm, a height of 4.25 cm, and a weight of only 185 g. It covers a distance of 16 m, recording one angle-distance pair every 0.36° of rotation.

The sensor was placed as in [19], i.e., before the knees, 80–100 cm apart (see Fig. 2(a)). The chair height was ergonomically adjusted for a $\sim 90^\circ$ angle between the worker's upper and lower legs. By gradually moving the chair away from the sensor, we determined that the maximum distance from which the sensor could still capture sufficiently sharp and sized leg outlines was ~ 2 meters. The worker could thus move up to one meter away from the desk (in addition to the 80–100 cm). This was in line with standard desk work, as people cannot work at a desk more than one meter away from it.

C. Leg Positions

The system recognizes the correct leg position according to the ISO 11226 standard, i.e., legs parallel and angle between upper and lower legs between $\sim 90^\circ$ and $\sim 135^\circ$ [25]. It also recognizes 14 incorrect positions that were considered as they frequently occurred while monitoring desk workers:

- left leg crossed over the right leg (and vice versa);
- left leg on the right leg's knee (and vice versa);
- legs extended;
- legs retracted;
- right leg extended forward with left leg put over the other, knee bent to a 90-degree angle (and vice versa);
- right leg crossed over the left leg, with both feet on the ground (and vice versa);
- right leg with knee bent to a 90-degree angle, foot on the ground, the other leg under the buttocks (and vice versa);
- left leg crossed over the right leg, with both legs on the chair (and vice versa).

With all these postures recognized, the system is significantly more specific than [19] as it helps identify which body parts are at risk, thereby improving the ergonomic risk assessment. Table I shows each leg position, its identifier (ID), and three LiDAR scans: *normal*, *noisy* due to motion, and *rotated*.

TABLE I
THREE PLOTS OF LIDAR SCAN FOR EACH LEG POSITION

ID	Name	Image	Normal	Noisy	Rotated
0	Correct				
1, 2	left leg crossed over the right leg				
3, 4	right leg on the left leg's knee				
5	legs extended				
6	legs retracted				
7, 8	right leg extended forward with left leg over the other knee bent 90 degrees				
9, 10	right leg crossed over the left leg, with both feet on the ground				
11, 12	right leg with knee bent 90 degrees, foot on the ground, and the other leg under the buttocks				
13, 14	left leg crossed over the right leg, with both legs on the chair				

D. Experimental Protocol

A Python application was developed to collect LiDAR data. The sensor frequency was set to 5 Hz, generating scans made up of ~ 1000 angle-distance pairs. Videos of the legs were only recorded to associate each LiDAR scan with the worker's leg position ID (label). Workers changed leg position every ~ 10 minutes: the sensor recorded many similar scans during those periods. To ensure a balanced dataset, 10% of similar scans per position were selected discarding the others, maintaining nuances. As Table I shows, the leg positions were easily detectable from the video frames. Scans were divided into three classes,

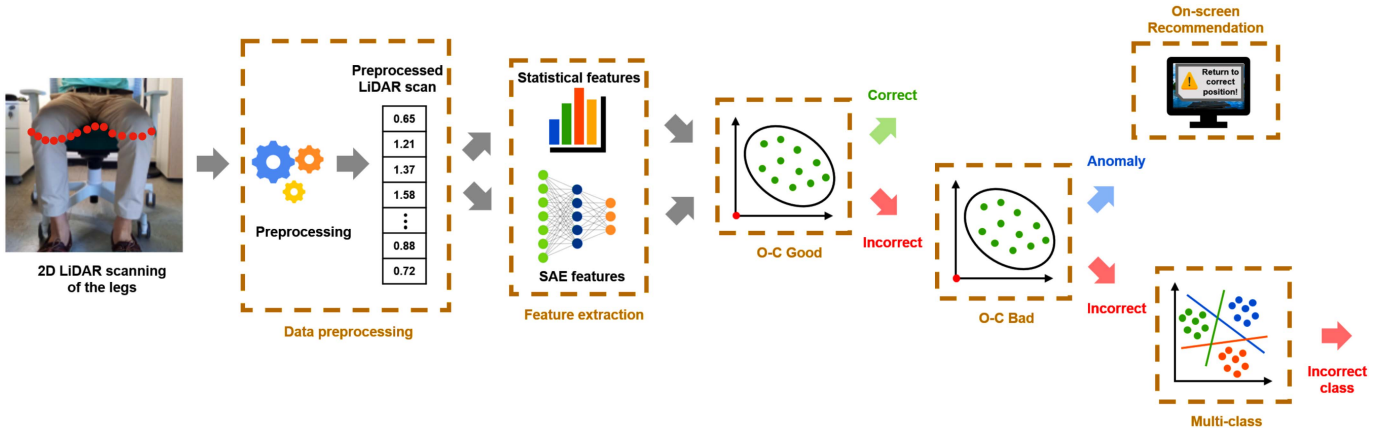


Fig. 3. System overview. The LiDAR scan is preprocessed, then features are extracted. *O-C Good* and *O-C Bad* recognize if the leg position is correct or incorrect, respectively, whereas *Multi-class* diversifies 14 incorrect positions.

TABLE II
PLOTS OF THE LiDAR SCANS OF THREE ANOMALIES

Anomaly 1	Scan 1	Anomaly 2	Scan 2	Anomaly 3	Scan 3

correct and common incorrect based on leg position IDs; the others were considered as anomalies.

The scans of correct position were 2467 (~91 per participant), and those of each incorrect position ranged from 314 to 393 (~161 per participant). A total of 4396 scans of incorrect positions were selected from the set, choosing 314 scans per position representing shades of that position. Uncommon incorrect positions were considered anomalies. Table II shows three anomalies; the others were chair rotated 90 degrees, and the presence of a colleague.

IV. SYSTEM

A. Overview

The system recognizes 15 leg positions: the *correct position*, 14 common incorrect positions, referred to as *incorrect positions* (see Table I), and *anomalies*, i.e., uncommon incorrect positions. If two alternatives of a given position can be obtained by exchanging the legs, Table I shows one whose ID is underlined: the other has a flipped scan.

As Fig. 3 shows, the system is made up of six modules:

- 1) *Data preprocessing*: obtains a preprocessed LiDAR scan (PLS) (see Section IV-B);
- 2) *Feature extraction*: extracts statistical features from the PLS and combines them with part of the latent space of a stacked autoencoder trained to reconstruct the PLS (see Section IV-C);
- 3) *O-C Good*: a one-class classifier that determines if the leg position is correct (see Table I);
- 4) *O-C Bad*: a one-class classifier that determines if the leg position is among those the system can recognize;

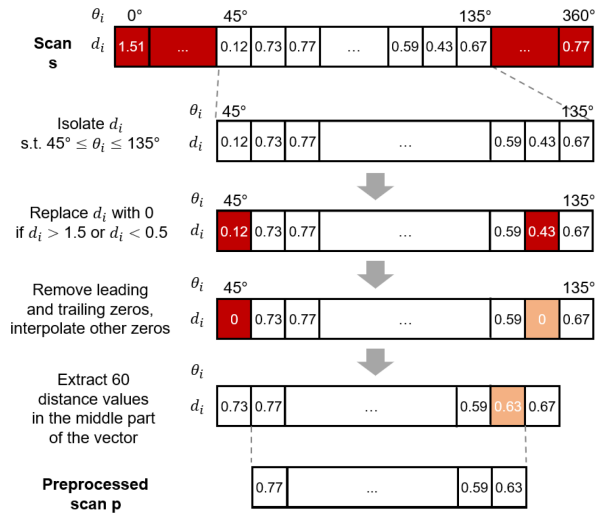


Fig. 4. Preprocessing of a LiDAR scan s . Red and orange squares denote the distance values removed and those obtained by interpolation.

- 5) *Multi-class*: a multi-class classifier that determines the incorrect position ID from 1 to 14 (see Table I);
- 6) *Recommendation*: a module that sends on-screen tips that suggest returning to the correct position.

The system's output is 0 if the leg position is correct, or an ID in $\{1, \dots, 14\}$ if the position is an incorrect position that *Multi-class* can recognize. If the system cannot recognize any position, it returns -1 (anomaly).

B. Data Preprocessing

LiDAR scans were preprocessed to isolate the points outlining the legs from those defining the contour of walls and furniture. Each LiDAR scan $s = ((\theta_1, d_1), \dots, (\theta_N, d_N))$ was preprocessed as follows (see Fig. 4):

- 1) angle-distance pairs (θ_i, d_i) with $\theta_i < 45^\circ$ or $\theta_i > 135^\circ$ were discarded as they outline objects beyond the desk;
- 2) angle-distance pairs (θ_i, d_i) with $d_i > 1.5$ or $d_i < 0.5$ were discarded as the sensor was before the knees, nearly one meter apart, with a minimum distance from the person to the sensor always exceeding 0.8 meters;

- 3) leading and trailing angle-distance pairs (θ_i, d_i) such that $d_i = 0$ (non-returning pulses) were discarded. The remaining $d_k = 0$ (transmission issues) were replaced with a linear interpolation based on the previous (d_{k-1}) and next (d_{k+1}) distance values;
- 4) the central scan area was chosen to reduce distance values, with 60 values being adequate to depict the legs.

Distance values were then rescaled in $[0, 1]$ using min-max normalization. Let $\mathcal{P}$ contain the preprocessed scans and let each preprocessed scan $\mathbf{p} \in \mathbb{R}_+^{M_p}$ be made up of M_p distance values p_j , where $j = 1, \dots, M_p$. Also, let the set of all the distance values be $\mathcal{L} = \bigcup_{\mathbf{p} \in \mathcal{P}} \bigcup_{j=1}^{M_p} p_j$. The normalized distance value n_j of p_j of each scan $\mathbf{p}$ was obtained as $n_j = \frac{p_j - \min\{\mathcal{L}\}}{\max\{\mathcal{L}\} - \min\{\mathcal{L}\}}$. Each scan generated a normalized scan $\mathbf{n} = (n_1, \dots, n_{M_p})$.

C. Features

Each normalized scan $\mathbf{n}_i^w$ of worker w was represented as a feature vector $\mathbf{f}_i^w$ comprising *statistical features* and *features extracted by a stacked autoencoder*.

1) **Statistical Features:** The statistical features (“Stats”) extracted from each normalized scan were as follows [19]: arithmetic, geometric and harmonic means, max, min, median, mode, max-to-min difference, std deviation, variance, kurtosis, skewness, number of peaks, mean height of the peaks, std deviation of the heights of the peaks, ratio of the highest absolute value to the root-mean-square value, root-sum-of-squares level, and root-mean-square value.

2) **Stacked Autoencoder Features:** Due to the higher variability in the scan shapes compared to [19], we considered an additional feature set (“SAE”) comprising part of the latent-space of a stacked autoencoder made up of two autoencoders (*AE1* and *AE2*). *AE1* was trained to reconstruct a normalized scan; *AE2* the latent representation of *AE1*. The MSE of both autoencoders was minimized (see Section II-B). Including features from a stacked autoencoder led to overcoming the limitations of [19], whose performance decreased when slightly rotating the chair or changing the worker-to-sensor distance. Stacked autoencoders can learn complex patterns and spatial relationships invariant to rotation, scale, and perspective changes [21]. This ensured consistent performance across different distances and orientations, improving robustness in real-world settings where variations are frequent.

D. Data Partitioning

In order to guarantee the independence of all the models from the particular worker, we generated w datasets $\mathcal{D}^w$, each associated with a worker w , where $w \in \{1, \dots, 27\}$. Each dataset $\mathcal{D}^w$ comprised a training (test) set $\mathcal{D}_{train}^w$ ($\mathcal{D}_{test}^w$) containing samples from all the other workers $u \neq w$ (of worker w only). These sets were as follows:

$$\mathcal{D}_{train}^w = \bigcup_{u \neq w} \bigcup_{i=1}^{N^u} (\mathbf{f}_i^u, ID_i^u), \quad \mathcal{D}_{test}^w = \bigcup_{i=1}^{N^w} (\mathbf{f}_i^w, ID_i^w) \quad (1)$$

where $\mathcal{D}_{train}^w$ and $\mathcal{D}_{test}^w$ contained $N_{train}^w = \sum_{u \neq w} N^u$ and $N_{test}^w = N^w$ samples, respectively. Each sample of the training set comprised a feature vector $\mathbf{f}_i^u$ from a normalized scan of

a worker $u \neq w$ and the corresponding leg position identifier ID_i^u from those in Table I. The test set samples had the same structure, but they all pertained to worker w . Subset $\mathcal{D}_{train}^w$ was used for feature selection, and to train various types of classifier. Subset $\mathcal{D}_{test}^w$ was used for testing.

E. Optimizing the Stacked Autoencoder

A grid search led to the best hyperparameter configuration of both AEs, i.e., a combination made up of the number of encoding neurons, and the types of activation functions of the encoding and decoding neurons. The grid search involved training the AEs using data partitioned as explained in Section IV-D. In particular, for each pair of worker w and hyperparameter configuration, the AEs were trained using the data in $\mathcal{D}_{train}^w$ and tested with those in $\mathcal{D}_{test}^w$.

The hyperparameter configuration that achieved the minimum average of the MSEs on the test sets $\mathcal{D}_{test}^w$ over all workers $w = 1, \dots, 27$ was considered the best. We tested *AE1* using encoding neurons from 2 to 30 in increments of 1. The logistic sigmoid and the positive saturating linear activation functions were tested for the encoding and decoding neurons. *AE1* achieved the lowest MSE using 20 encoding neurons, and the logistic sigmoid and linear activation functions for the input and output neurons, respectively.

AE2 was trained varying the number of encoding neurons from 2 to 20 in increments of 1. *AE2* obtained the lowest MSE when using 3 encoding neurons based on the logistic sigmoid activation function, and decoding neurons based on a linear activation function. The *AE2* latent-space representation (3 features) was combined with the statistical features, thus obtaining 21 features.

F. Feature Selection

The *sequential feature selection* (SFS) was used to reduce features [26]. The levels of accuracy of an O-C SVM and a multi-class SVM were used to select the best features for *O-C Good* and *O-C Bad*, and *Multi-class*, respectively, trained with data partitioned as explained in Section IV-D. For *O-C Good* (*O-C Bad*), the training sets only contained samples of correct (incorrect) position, whereas the test sets had a balanced number of samples of correct (incorrect) position and incorrect positions (anomalies).

In particular, *O-C Good* was trained using 2376 samples, whereas the test sets contained 182 samples. *O-C Bad* was trained using 4234 samples, and tested using 182 samples. For *Multi-class*, training and test sets preserved the percentage of samples for each class of the original dataset. The mean accuracy on the test sets (ratio of correctly classified samples to the total number of samples) was maximized.

The most selected features throughout 30 independent SFS executions included the arithmetic and harmonic means, and kurtosis for all the three classifiers. Features were as follows:

- *O-C Good*: median, skewness, the ratio of the highest absolute value to the root-mean-square value, root-sum-of-squares level, and two SAE features;
- *O-C Bad*: max-to-min difference, number of peaks, mean height of the peaks, and all SAE features;

TABLE III
MEAN LEVELS OF ACCURACY AND CORRESPONDING STANDARD DEVIATIONS ON THE TEST SETS OF THE TESTED CLASSIFIERS

Features	O-C Task	One-Class (O-C) Classifiers				Features	Multi-Class Classifiers			
		O-C SVM	SVDD	IF	LOF		SVM	k-NN	MLP	DT
Stats	O-C Good	0.957 ± 0.012	0.897 ± 0.010	0.633 ± 0.016	0.564 ± 0.012	Stats	0.970 ± 0.011	0.949 ± 0.015	0.932 ± 0.013	0.682 ± 0.012
	O-C Bad	0.867 ± 0.011	0.908 ± 0.013	0.667 ± 0.015	0.573 ± 0.009					
SAE	O-C Good	0.939 ± 0.006	0.879 ± 0.010	0.598 ± 0.013	0.527 ± 0.016	SAE	0.958 ± 0.004	0.886 ± 0.016	0.876 ± 0.011	0.853 ± 0.012
	O-C Bad	0.874 ± 0.009	0.917 ± 0.014	0.661 ± 0.017	0.564 ± 0.010					
Stats ∪ SAE	O-C Good	0.976 ± 0.005	0.929 ± 0.012	0.657 ± 0.011	0.549 ± 0.009	Stats ∪ SAE	0.986 ± 0.008	0.965 ± 0.006	0.965 ± 0.011	0.732 ± 0.014
	O-C Bad	0.989 ± 0.005	0.926 ± 0.011	0.647 ± 0.016	0.562 ± 0.013					

The best values are in bold.

- *Multi-class*: median, skewness, number of peaks, the ratio of the highest absolute value to the root-mean-square value, root-sum-of-squares level, and all SAE features.

These features were used to implement the classifiers.

G. O-C Good and O-C Bad One-Class Classifiers

O-C Good and *O-C Bad* were implemented as two one-class SVMs whose hyperparameters were found as follows. Linear, polynomial, and Gaussian kernels were tested, varying γ from 0.1 to 1, in increments of 0.01, to control the kernel width. O-C SVMs were trained 27 times, each time using the training $\mathcal{D}_{train}^w$ and test $\mathcal{D}_{test}^w$ sets of worker w . We selected the best hyperparameter configuration based on the mean accuracy over all workers' test sets. *O-C Good* (*O-C Bad*) achieved the highest accuracy using the Gaussian kernel, and γ equal to 0.8 (0.65).

H. Multi-Class Classifier

SVMs were tested using linear, polynomial, and Gaussian kernels with values of γ ranging from 0.1 to 1 in increments of 0.01, using the One-vs-One and One-vs-All approaches, considering 14 classes (all the incorrect positions).

In the One-vs-All approach, one binary SVM per class was trained to distinguish that class from the others (14 SVMs in total). The One-vs-One approach required the training of 91 (i.e., $\binom{14}{2}$) binary SVMs, each aimed at distinguishing between two classes. One-vs-All and One-vs-One SVMs were trained and tested 27 times, each time using the training $\mathcal{D}_{train}^w$ and test $\mathcal{D}_{test}^w$ sets of worker w (see Section IV-D). Regarding the One-vs-One approach, each binary SVM was trained using the samples associated with the two classes involved. Based on the mean accuracy on all workers' test sets (99.2%), the Gaussian kernel with $\gamma = 0.95$ was the best.

I. System's Accuracy

The system's accuracy was assessed by inputting data from $\mathcal{D}_{test}^w$ and calculating the ratio of correctly classified samples to the total number of samples for each worker w . This process was repeated for each of the 27 workers ($w = 1, \dots, 27$), and the results are presented in Table IV. The consistent accuracy levels observed across workers (standard deviation of only 5.4%) highlight the system's generalization capability and its independence from specific worker characteristics.

J. Recommendation

This module monitors workers' leg posture and displays a pop-up on their computer screen if they sit incorrectly for too long. Pop-ups suggest returning to the correct position.

TABLE IV
SYSTEM'S ACCURACY FOR EACH WORKER w WHEN TESTED ON $\mathcal{D}_{test}^w$

Worker w	Accuracy	Worker w	Accuracy	Worker w	Accuracy
1	98.41%	10	97.61%	19	98.01%
2	98.80%	11	98.41%	20	98.41%
3	97.61%	12	98.01%	21	98.01%
4	98.41%	13	97.61%	22	97.22%
5	98.01%	14	99.20%	23	98.01%
6	97.80%	15	97.61%	24	98.80%
7	98.41%	16	97.22%	25	98.80%
8	97.61%	17	97.61%	26	98.01%
9	98.41%	18	99.20%	27	98.01%

The maximum time a worker can sit with incorrect leg positions can be set based on advice from experts, based on existing problems, such as pain and musculoskeletal disorders.

The system saves the time and response to each pop-up, recording whether the worker followed or ignored the advice. The system generates a diagram (called *advanced timeline*) that displays this information over time. This helps safety engineers identify which workers benefit from the recommendation system and which, instead, need training or assistance as they tend to ignore notifications. Advanced timelines are discussed in the final test (Section VI).

V. COMPARISON WITH OTHER CLASSIFIERS

This section compares the accuracy of the system's classifiers to other classifiers. Table III shows the mean accuracy and standard deviation of each classifier, trained using the "Stats", "SAE", and "Stats ∪ SAE" feature sets. Hyperparameters were optimized by repeating the training 27 times, each time using the dataset $\mathcal{D}^w$ of worker w .

A. One-Class Classification

O-C Good and *O-C Bad* were compared to the Support Vector Data Description (SVDD) [23], Isolation Forest (IF) [27], and Local Outlier Factor (LOF). These models were chosen as they are efficient for one-class classification [28].

a) *Support vector data description (SVDD)*: Various models were trained, varying the kernel parameter γ from 0.1 to 1, in increments of 0.01, using the linear, quadratic, and Gaussian kernels. The highest accuracy (93.1%) was achieved using Stats ∪ SAE, and a Gaussian kernel with $\gamma = 0.79$.

b) *Isolation forest (IF)*: An IF makes a random sampling of data and arranges them in a tree structure based on randomly selected features. The fewer the nodes a sample passes through, the more likely the sample is an outlier, as the tree easily separates it from the other observations. Various IF classifiers

were trained, varying the number of trees from 5 to 15. The highest accuracy (67.2%) was obtained using 10 trees and Stats $\cup$ SAE.

c) *Local outlier factor (LOF)*: The local outlier factor (LOF) measures the local density deviation of a data point w.r.t. its neighbors to detect outliers. The LOF score of a data point is the ratio of the average local density of its k nearest neighbors to the local density of the data point. The higher the score, the more likely the data point is an outlier. LOF was trained using the Euclidean, Mahalanobis, and Minkowski distances, varying k from 5 to 15. The Euclidean distance, $k = 10$, and Stats $\cup$ SAE led to the best accuracy (58.1%).

B. Multi-Class Classification

Multi-class was compared to the k -nearest neighbor [29], multi-layer perceptron, and decision tree [30], [31].

a) *k-NN*: The k -nearest neighbor (k -NN) finds the k data points closest to an input data point. The output is calculated by aggregating the output values of the k nearest neighbors, also using weighted voting. Equal and inverse weights, Euclidean and Manhattan distances, and $k \in \{1, \dots, 6\}$ were used to train the k -NN. Inverse weights, Euclidean distance, and $k = 1$ led to the highest mean accuracy (97.8%).

b) *Multi-layer perceptron (MLP)*: One hidden layer and a number of neurons in $\{3, \dots, 15\}$ were tested for the MLPs. More neurons or hidden layers did not improve performance. MLPs were trained using the scaled conjugate gradient, Levenberg–Marquardt, and Bayesian regularization algorithms. The highest mean accuracy (95.4%) was achieved with 9 hidden neurons and the Levenberg–Marquardt algorithm.

c) *Decision tree (DT)*: Gini criterion was used to train the DTs. The max tree depth d , max number of splits s , min number of leaf nodes l , and number of features f were varied such that $d \in \{5, \dots, 15\}$, $s \in \{3, \dots, 8\}$, $l \in \{3, \dots, 8\}$, and $f \in \{1, \dots, 6\}$, respectively. The highest mean accuracy (89.5%) was achieved with $d = 6$, $s = 7$, $l = 3$, and $f = 4$.

VI. DEMONSTRATION

A. Setup

The system was tested involving 3 workers (testers) who did not take part in the training phase: worker 1, a man wearing trousers, and workers 2 and 3, two women wearing trousers and a skirt, respectively. Testers were selected to obtain a uniform height and weight distribution: they were 1.86 m, 1.67 m, and 1.58 m tall, weighing 94.7 kg, 72.3 kg, and 53.2 kg, respectively. Testers' legs were video recorded to label the scans. Videos showed that testers maintained their leg positions for $\sim 2'47''$, on average; we thus set a scanning frequency of 2' to capture the most changes in leg position. Data were collected for 6 hours, 180 samples per tester.

B. Classification Accuracy

The classifiers achieved the accuracy, precision, and recall levels summarized in Table V. As Fig. 5 shows, *O-C Good*, *O-C Bad*, and *Multi-class* misclassified four, two, and four samples,

TABLE V
PERFORMANCE OF EACH CLASSIFIER DURING THE DEMONSTRATION

Worker	Accuracy			Precision			Recall		
	1	2	3	1	2	3	1	2	3
O-C Good	99.4%	99.4%	98.9%	98.6%	100%	98.0%	100%	98.6%	98.0%
O-C Bad	99.1%	100%	99.2%	99.0%	100%	100%	100%	100%	99.2%
Multi-class	99.0%	99.1%	98.4%	99.2%	99.2%	98.9%	99.2%	99.2%	98.7%

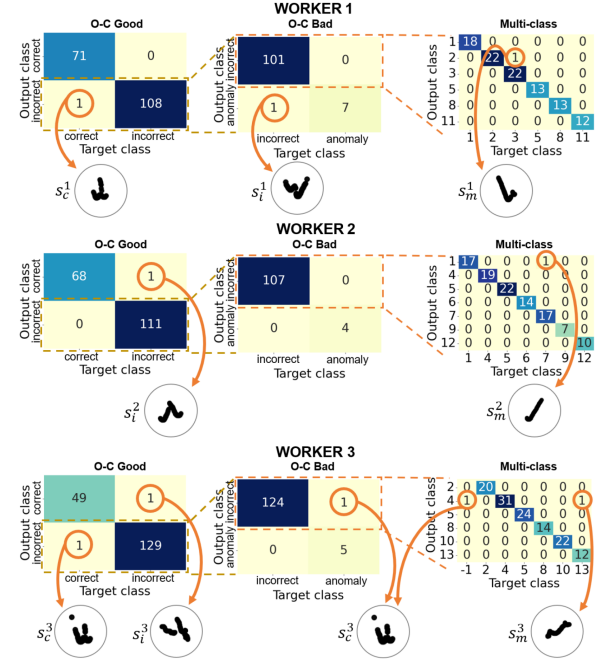


Fig. 5. Confusion matrices of the O-C Good (left), O-C Bad (center), and Multi-class SVMs (right) classifiers of worker 1, 2, and 3.

respectively, whose scans are under the corresponding confusion matrix. Watching the video of the legs revealed that these errors occurred with participants in motion.

Regarding worker 1 (top of Fig. 5), *O-C Good* and *O-C Bad* misclassified only one sample; *Multi-class* confused incorrect positions 2 and 3 with each other. The videos revealed that:

- the sample misclassified by *O-C Good* corresponded to a scan (s_c^1 in Fig. 5) recorded with the chair rotated;
- the sample misclassified by *O-C Bad* was due to scanning a change in position (s_t^1 in Fig. 5);
- the sample misclassified by *Multi-class* as right leg crossed over the left leg came from a scan (s_m^1 in Fig. 5) of the right leg on the left leg's knee: the worker was rotating the chair.

Regarding worker 2 (middle of Fig. 5), *O-C Good* and *Multi-class* misclassified one sample of incorrect position, and one sample of legs crossed as another way of crossing legs, respectively. The video showed that these errors occurred as the worker was scratching his belly.

Concerning worker 3, *O-C Good* misclassified one sample of correct position and another of incorrect position. Again, the video showed that the scans (s_c^3 and s_t^3 in Fig. 5) were recorded while rotating the chair. The sample derived from s_c^3 was also misclassified by *O-C Bad*. Finally, *Multi-class* misclassified

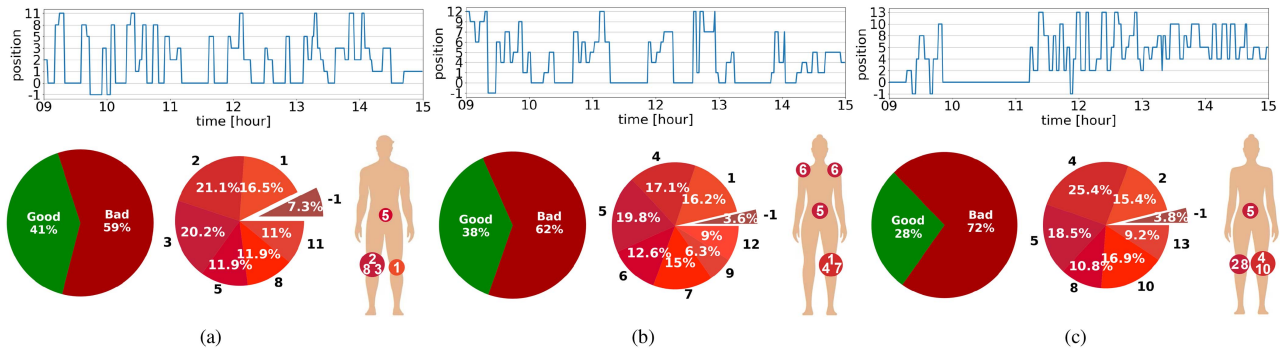


Fig. 6. Leg position timelines of testers 1 (a), 2 (b), and 3 (c) during six hours (top), and corresponding pie charts and body part risk plots (bottom).

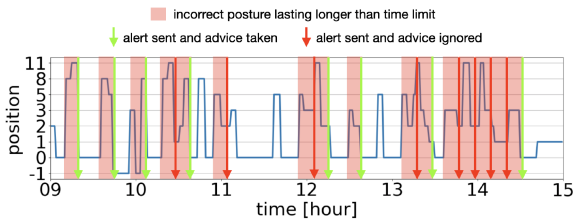


Fig. 7. Advanced timeline of worker 1.

one sample of both legs on the chair (position 13) as right leg crossed over the other (position 4). The video showed that worker 3 was standing and leaning forward. Sample s_c^3 was also misclassified by *Multi-class* as it mistakenly received in input an anomaly due to an error of *O-C Bad: Multi-class* cannot classify anomalies.

C. Dashboard

The system provides safety engineers with an innovative dashboard based on *timelines*, *pie charts*, and *silhouettes*. This dashboard includes more detailed information than [19], such as at-risk body parts, the entity of this risk, the leg positions stressing each body part, the percentage of time spent in good/bad postures, and each worker's tendency to take/ignore posture recommendations. All this additional information leads to assessing the workers' risk more precisely, thereby reducing the risk of musculoskeletal disorders and pain more effectively.

This section describes a dashboard presenting daily leg position statistics, as an example. For each worker, the timeline shows the leg positions over time. Two pie charts show the percentage of time spent in good and bad positions (*Good/Bad pie chart*), and in each incorrect position, respectively. A silhouette shows at-risk body parts based on the most long-lasting incorrect positions. The maximum duration for incorrect positions was set to 10 min based on advice from experts.

1) Timelines: The top of Fig. 6 shows the leg position timelines of workers 1, 2, and 3. As can be seen, worker 1 changed various incorrect positions (2, 3, 5, 8, and 11) with the correct position and some anomalies. Also, he frequently changed position, thus reducing the load on muscles and joints, decreasing the risk of health problems [25]. Fig. 7 shows the advanced timeline for worker 1. Each red stripe is a 10-min interval where the

worker had his legs in an incorrect position. After each interval, the system sent a recommendation encouraging a change of position: accepted and ignored recommendations are indicated by green and red arrows, respectively. As Fig. 7 shows, the worker accepted the first three recommendations, ignored the fourth one (at $\sim 10:30$), and accepted the next one, maintaining the correct leg position for the next half-hour, with only 4 min spent with one leg on top of the other, below the knee (position 8, Table I). In the afternoon, he ignored four consecutive notifications from 13:30, keeping incorrect positions for ~ 50 min, maybe due to annoyance.

2) Pie Charts: Regarding worker 1, the Good/Bad pie chart shows that he sat for 59% of the time (over 4 out of 6 working hours) with incorrect leg positions. Positions 2 and 3, two ways of crossing legs (see Table I), were the most frequent: they were maintained, in total, for over two and a half hours per day, 21.1% and 20.2% of the time, respectively.

Workers 2 and 3 kept bad positions longer than worker 1, with 62% and 72% of their time spent in incorrect positions, respectively: an increase of about 11 and 47 min. In particular, worker 2 spent about one hour in position 5, which puts excessive weight on the lower back, increasing the risk of pain [32]. Positions 1 and 4 (crossed legs) were also frequent for worker 2, accounting for 17.1% and 16.2% of the time, respectively: almost two hours (33.3%) in total. Crossed legs increase the risk of circulatory issues and pelvic tilt [33].

Worker 3 spent almost 4 hours (72%) sitting with incorrect positions: the longest of the three workers. She preferred position 4 (crossed legs), which leads to a bent and asymmetrical posture, thereby increasing the risk of back pain [33]. The timeline shows that she ignored all the recommendations after 11 a.m. This may be due to her young age (25), which correlates with a low risk perception [34].

Thus, the system can identify which workers follow the recommendations and which do not, helping to estimate risk perception and identify the body parts at risk. Safety engineers can intervene with personalized safety training, maximizing persuasiveness and reducing risk.

3) Silhouettes: Each worker's silhouette indicates at-risk body parts using circles: the higher the risk, the wider the circle. The risk of body part b is higher [25]:

- the longer the total time a worker spends with incorrect positions that stress body part b (*stressful positions*);

- the shorter the time the worker spends, on average, with non-stressful leg positions for body part b between two periods of stressful positions for that body part.

Each circle contains the corresponding stressful position IDs.

The silhouettes in Fig. 6 show that all workers' knees are at risk, left knee for worker 1, right knee for workers 2 and 3. The stressful positions that put worker 1's left knee at risk are positions 2, 3, and 8, i.e., three ways of crossing the legs (see Table I). As the pie chart shows, worker 1 maintained these positions for a total percentage of time of 53.2% (over 3 hours). Worker 1's right knee is at low risk as he stayed in the related stressful position (ID=1) for only 16.5% of the time, less than one hour. The silhouette also shows that worker 1's back is at risk due to position 5 (legs extended), maintained for 11.9% of the time. This causes the worker to shift his weight forward and arch the back, overloading the lumbar area.

The silhouettes of workers 2 and 3 highlight that the right knee is at risk as worker 2 (3) sits with positions 1, 4, and 7 (4 and 10) for 48.3% (42.3%) of the workday, respectively. These positions involve crossing the left leg over the right leg, putting strain on the right knee. The back is another body part at risk for workers 2 and 3, although to a lesser extent than the knees, as the circle sizes show. Also, worker 2's shoulders are stressed due to position 6 (legs retracted), as the worker rests on her elbows on the desk. In the medium-long term, position 6 can lead to shoulder impingement syndrome, with pain, weakness, and loss of movement at the shoulder.

As shown, silhouette analysis is key for safety engineers to evaluate the risk to workers' body parts. The risk is reduced if workers follow the recommendations. Those who tend to ignore recommendations (like worker 3) could be grouped based on at-risk body parts and provided with tailored safety training to correct bad habits.

VII. CONCLUSION

This paper has presented a system that determines 15 leg positions of desk workers with over 98% accuracy, while also rejecting unrecognized positions. Workers' privacy is respected by determining the leg positions using 2D LiDAR scans, which produce raw curves outlining the legs that cannot unveil any sensitive information.

The system helps improve posture by sending workers real-time recommendations. It also supports safety engineers in determining which workers are at risk and which body parts are the most stressed by long-lasting incorrect positions.

Future work will aim to personalize recommendations based on the worker's behavior and use LiDAR sensors to detect upper-body posture, thereby maximizing workers' well-being in years to come.

REFERENCES

- [1] World Health Organization, "WHO's work on physical activity," 2020.
- [2] J. P. Caneiro et al., "The influence of different sitting postures on head/neck posture and muscle activity," *Manual Ther.*, vol. 15, no. 1, pp. 54–60, 2010.
- [3] E. Grandjean and W. Hünig, "Ergonomics of posture—review of various problems of standing and sitting posture," *Appl. Ergonom.*, vol. 8, no. 3, pp. 135–140, 1977.
- [4] N. Bloom et al., "Does Working from Home Work? Evidence from a Chinese Experiment," *Q. J. Econ.*, vol. 130, no. 1, pp. 165–218, 2015.
- [5] P. Markopoulos et al., "Neckio: Motivating neck exercises in computer workers," *Sensors*, vol. 20, no. 17, 2020, Art. no. 4928.
- [6] Q. Hu et al., "A smart chair sitting posture recognition system using flex sensors and FPGA implemented artificial neural network," *IEEE Sensors J.*, vol. 20, no. 14, pp. 8007–8016, Jul. 2020.
- [7] O. Krejcar et al., "A smart system for sitting posture detection based on force sensors and mobile application," *Mobile Inform. Syst.*, no. 1, 2020, Art. no. 6625797. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2020/6625797>
- [8] A. R. Anwary et al., "Smart-cover: A real time sitting posture monitoring system," *Sensors Actuator A, Phys.*, vol. 317, 2021, Art. no. 112451.
- [9] S. Muppavram, N. Patel, and M. Nadeem, "Posture alert," in *Proc. IEEE Region Ten Symp.*, 2018, pp. 213–218.
- [10] C. Ma et al., "Activity level assessment using a smart cushion for people with a sedentary lifestyle," *Sensors*, vol. 17, no. 10, 2017, Art. no. 2269.
- [11] J. Wang, B. Hafidh, H. Dong, and A. E. Saddik, "Sitting posture recognition using a spiking neural network," *IEEE Sensors J.*, vol. 21, no. 2, pp. 1779–1786, Jan. 2021.
- [12] H. Jeong and W. Park, "Developing and evaluating a mixed sensor smart chair system for real-time posture classification: Combining pressure and distance sensors," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 5, pp. 1805–1813, May 2021.
- [13] K. R. Kini, F. Harrou, M. Madakyaru, F. Kadri, and Y. Sun, "Automatic detection of unbalanced sitting postures in wheelchairs using unlabeled sensor data," *IEEE Sensors Lett.*, vol. 6, no. 8, Aug. 2022, Art. no. 6002704.
- [14] P. Vermader, A. Mancisidor, I. Cabanes, N. Perez, and J. Torres-Unda, "Intelligent sitting posture classifier for wheelchair users," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 944–953, 2023.
- [15] E. Rocha-Ibarra et al., "Kinect validation of ergonomics in human pick and place activities through lateral automatic posture detection," *IEEE Access*, vol. 9, pp. 109067–109079, 2021.
- [16] E. S. Ho et al., "Improving posture classification accuracy for depth sensor-based human activity monitoring in smart environments," *Comput. Vis. Image Understanding*, vol. 148, pp. 97–110, 2016.
- [17] C.-W. Chen, T. Määttä, Kevin Bing-Yung Wong, and H. Aghajan, "A collaborative framework for ergonomic feedback using smart cameras," in *Proc. ACM/IEEE Int. Conf. Distrib. Smart Cameras*, 2012, pp. 1–6.
- [18] B. Pollard et al., "Movement at work: A comparison of real time location system, accelerometer and observational data from an office work environment," *Appl. Ergonom.*, vol. 92, 2021, Art. no. 103341.
- [19] F. Pistolesi, M. Baldassini, and B. Lazzarini, "Are you sitting with good posture? Tracking the position of the legs via 2D LiDAR," *IEEE Embed. Syst. Lett.*, vol. 15, no. 3, pp. 137–140, Sep. 2023.
- [20] R. Roriz, J. Cabral, and T. Gomes, "Automotive LiDAR technology: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6282–6297, Jul. 2022.
- [21] J. Zhai, S. Zhang, J. Chen, and Q. He, "Autoencoder and its various variants," in *Proc. IEEE Int. Conf. Syst. Man Cybern.*, 2018, pp. 415–419.
- [22] B. Schölkopf et al., "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, 2001.
- [23] D. M. Tax and R. P. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, 2004.
- [24] Q. Wang, "Support vector machine algorithm in machine learning," in *Proc. IEEE Int. Conf. Artif. Intell. Comput. Appl.*, 2022, pp. 750–756.
- [25] E. Commission et al., "ISO 11226, Ergonomics – Evaluation of static working postures," 2000.
- [26] I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *J. Mach. Learn. Res.*, vol. 3, pp. 1157–1182, 2003.
- [27] F. T. Liu et al., "Isolation-based anomaly detection," *ACM Trans. Knowl. Discov. Data*, vol. 6, no. 1, pp. 1–39, 2012.
- [28] S. S. Khan and M. G. Madden, "A survey of recent trends in one class classification," in *Proc. Artif. Intell. Cogn. Sci.*, 2010, pp. 188–197.
- [29] F. Lalonde and K. Doya, "Numerical data imputation: Choose kNN over deep learning," in *Proc. Similarity Search Appl.*, 2022, pp. 3–10.
- [30] S. B. Kotsiantis, "Decision trees: A recent overview," *Artif. Intell. Rev.*, vol. 39, no. 4, pp. 261–283, 2011.
- [31] S. Amendolia et al., "A comparative study of k-nearest neighbour, support vector machine and multi-layer perceptron for thalassemia screening," *Chemometr. Intell. Lab. Syst.*, vol. 69, no. 1/2, pp. 13–20, 2003.
- [32] A. M. Lis et al., "Association between sitting and occupational LBP," *Eur. Spine J.*, vol. 16, no. 2, pp. 283–298, 2006.
- [33] K. Jung et al., "The effects of cross-legged sitting on the trunk and pelvic angles and gluteal pressure in people with and without low back pain," *Int. J. Environ. Health Res.*, vol. 17, no. 13, 2020, Art. no. 4621.
- [34] P. Slovic, "Perception of risk," *Science*, vol. 236, no. 4799, pp. 280–285, 1987.